什么是 encapsulation(封装)
什么是 polymorphism（多态）
什么是 Garbage Collection，GC（垃圾回收）

```
Integer i2 = 0;            //  自动装箱：Integer.valueOf(0)
int i1 = i2;               //  自动拆箱：i2.intValue()
```

GUI：
THREE concept：Component, Container, Event.
Component: An object that the user can see on the screen and can also interact with.
Container: A component that can hold other components.
Event: An action triggered by the user (e.g. pressing a key, clicking a mouse button).

如何设计 GUI
Designing a GUI involves creating components, putting them into containers, and arranging for the program to respond to events (e.g.by responding to mouse clicks).

一个例子

```
import javax.swing.*;
import java.awt.event.*;
public class simpleGUI{
    JButton button1;
    JButton button2;
    public static void main(String[] args){
    simpleGUI myGUI = new simpleGUI();
    myGUI.go();
    }
    public void go(){
    JFrame frame = new JFrame();
    this.button1 = new button(“Hello from button 1”);
    this.button2 = new button(“Hello from button 2”);
    button1.addActionListener(new button1Listener());
    button2.addActionListener(new button2Listener());
    frame.setLayout(new BorderLayout());
    frame.add(button1, BorderLayout.NORTH);
    frame.add(button2, BorderLayout.CENTER);
    frame.setSize(100,100)
    frame.setvisible(true);
    }
    // 第一个按钮的监听器
    class Button1Listener implements ActionListener {
        public void actionPerformed(ActionEvent e) {//注意有个 ed
            button1.setText("Hello!");
        }
```

```
    }

    // 第二个按钮的监听器
    class Button2Listener implements ActionListener {//注意一定要写对接口名字
        public void actionPerformed(ActionEvent e) {
            button2.setText("Bye!");
        }
    }
}
```

对应步骤

Describe the three steps required to implement an event handler for the button.

Step 1: Implement the ActionListener interface

In the declaration for the event handler class, code specifies that the class either:

- implements a listener interface,
- or extends a class that implements a listener interface.

For example:

Class button1Listener implements ActionListener

Step 2: Register with the widget

Code indicates that your program wants to listen for events by:

- registering an instance of the event handler class as a listener
- upon one or more components.

For example:

Button2.addActionListener(new button2Listener());

Step 3: Define the event-handling method

Code implements the methods in the listener interface,
typically by providing logic in the actionPerformed method to respond to the event.

For example:

```
Public void actionPerformed(ActionEvent e)
{
    //…
}
```

手写 java 程序

```
Import java.util.*
/**
 *这是一个测试类
 *为了帮助同学们快速梳理实践知识
 *
 *@author Tang
 *@version 1.0
 */
public class test{
```

```
    public int test_int = 10；
    public String[] test_String_Array = new String[3];
    public String test_String = “Hello”;
    public int[][] matrix= new int[1][1];
    public ArrayList<Integer> test_AList= new ArrayList<Integer>;

    /**
     *构造方法
     *
     *@param a  一个整数
     *@param b  一个整数
     */
    public test(int a,int b)
    {
        //Code…
    }
    /**
     *这是一个 setter 方法
     *
     *@param a  从外部接受的参数
     *@throws IllegalArgumentException 如果参数为 0，则会抛出异常
     */

    Public void setter_for_test_int(int a)
    {
        if( a == 0)
            throw IllegalArgumentException(“变量不能为 0!”);
        else
            Self.test_int = a;
    }


}
```

关于生成 javadoc

Javadoc -d doc Example.java

构造自己的异常

```
public class myException extends Exception{
    public myException(String message)
    {
        super(message);
    }
```

}

数据流（Stream）

是和数据的连接

包括 character stream，byte stream

也可分为 input 与 output

I/O 读取/输出

```
import java.io.*;
public class file_writter{
	public static void main(String[] args)
	{
		String fileName= "doc.txt";//待写入的文件
		File file = new File(fileName);
		If(file.exists())
		{
			System.out.println("文件已存在");
			System.exit(1);//异常退出
		}
		try {
			//如果是写入文件
			//从开始写或者读的时候就要进入 try，因为容易出错
			FileWriter fileWriter = new FileWriter(file);
			BufferedWriter bufferedwriter = new BufferedWriter(fileWriter);
			for(int I = 0;i<5;i++)
			{
				bufferedwriter.write(Integer.toString(i));//注意转成字符输入
				bufferedwriter.write("\n");
			}
			bufferedwriter.close();
			fileWriter.close();//先 b 后 f
			//如果是读取文件
			FileReader filereader = new FileReader(file);
			BufferedReader bufferedreader = new BufferedReader(filereader);
			String line = null;
			while((line = bufferedreader.readLine)!=null)
			{
				//do whatever you want to do with line…
			}
			bufferedreader.close();
			filereader.close();
		}
		catch(IOException e){
```

```
                e.printStackTrace();
            }
        }
    }
```